

Financially Motivated Cyber Operations: A Case Study of FIN7

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Abstract

This paper examines the financially motivated cyber operations of the threat group FIN7 and evaluates whether its activities align with the phases of a well-organized cyber operation. FIN7 is a sophisticated cybercriminal group known for targeting businesses in the retail, hospitality, and financial sectors to steal payment card data. Using a case study approach and document analysis of publicly available sources such as the MITRE ATT&CK framework and cybersecurity reports, this study analyzes FIN7's tactics, techniques, and procedures. The research applies the seven phases of a well-organized cyber operation: target identification, reconnaissance, gaining access, hiding presence, establishing persistence, execution, and assessment. Findings indicate that FIN7 follows a structured and methodical approach similar to traditional Advanced Persistent Threat groups, despite being financially motivated rather than state-sponsored. The group demonstrates clear objectives at each phase, ultimately aiming to maximize financial gain through data exfiltration and system compromise. These results suggest that financially motivated threat actors can operate with the same level of organization and sophistication as nation-state APT groups.

Introduction

Advanced Persistent Threat (APT) groups are often associated with nation-state cyber espionage; however, some cybercriminal organizations demonstrate similar levels of sophistication and operational structure while being primarily financially motivated. One such group is FIN7, a highly organized cybercrime group known for targeting businesses to steal financial and payment card data.

FIN7 has been responsible for numerous high-profile cyber incidents affecting retail, hospitality, and financial organizations. These attacks commonly involve phishing campaigns, malware deployment, and long-term persistence within victim networks. Public reporting has shown that FIN7 continues to evolve its operations, using fake cybersecurity companies and phishing techniques to deceive victims and gain access to systems [1].

The evolution of cybercrime has led to the emergence of highly organized groups that operate similarly to legitimate businesses. FIN7, in particular, has demonstrated a structured operational

model that includes defined roles, coordinated campaigns, and the use of front companies to conceal its activities. This level of organization challenges traditional distinctions between cybercriminal groups and Advanced Persistent Threat actors, as both now exhibit similar levels of planning, execution, and adaptability. Understanding this shift is critical for organizations seeking to defend against modern cyber threats.

The increasing frequency of financially motivated cyberattacks highlights the importance of understanding how these groups operate. While traditional APT groups are often driven by geopolitical objectives, groups like FIN7 demonstrate that financial gain can drive equally sophisticated and structured cyber operations.

This study examines the motivations that prompt FIN7 to conduct cyber operations and analyzes how the group's activities align with the phases of a well-organized cyber operation. The goal is to determine whether financially motivated threat actors follow a structured operational model and to identify the objectives associated with each phase.

Literature Review

Advanced Persistent Threat (APT) groups, also referred to as threat actors or activity groups, are clusters of malicious cyber activity tracked by cybersecurity organizations based on observed behavior and techniques [2]. These groups are commonly analyzed using frameworks such as the MITRE ATT&CK database, which maps adversary tactics, techniques, and procedures (TTPs) based on publicly reported data.

FIN7 is a financially motivated cybercriminal group known for conducting sophisticated cyber operations targeting organizations in the retail, hospitality, and financial sectors. According to MITRE ATT&CK, FIN7 has been active since 2013 and has targeted a wide range of industries, including retail, hospitality, financial services, and healthcare (Table 1) [2]. The group is known for using phishing emails to gain initial access, followed by the deployment of malware and the use of legitimate administrative tools to maintain persistence within victim networks.

Research on FIN7 indicates that the group operates with a high level of organization, often resembling the structure and methodology of nation-state APT groups. Their operations typically involve multiple stages, including reconnaissance, initial compromise, lateral movement, and data exfiltration. Additionally, FIN7 has demonstrated the ability to adapt its techniques over time, incorporating new tools and strategies to evade detection and improve operational success.

While existing research provides insight into the tools and techniques used by FIN7, there is less emphasis on analyzing their activities through the lens of a structured cyber operation model. Specifically, there is a gap in understanding how financially motivated groups like FIN7 align with the defined phases of a well-organized cyber operation.

Additional studies have highlighted the increasing convergence between financially motivated cybercrime groups and traditional APT actors. FIN7, in particular, has been noted for its structured operations, use of advanced malware, and ability to sustain long-term campaigns.

These characteristics suggest that the distinction between cybercriminal organizations and nation-state actors is becoming less defined, as both groups adopt similar tactics and operational frameworks.

Despite the availability of detailed threat intelligence on FIN7, much of the existing work focuses primarily on the group's tools and techniques rather than its operational structure. While frameworks such as MITRE ATT&CK provide valuable insight into specific tactics, they do not always fully capture how these tactics are organized into a complete operational lifecycle.

Additionally, prior research tends to emphasize nation-state Advanced Persistent Threat groups when discussing structured cyber operations, often overlooking financially motivated actors. This creates a gap in understanding whether cybercriminal organizations like FIN7 follow the same systematic processes as state-sponsored groups.

Addressing this gap is important because financially motivated groups are responsible for a significant portion of real-world cyber incidents. By analyzing FIN7 through the lens of a well-organized cyber operation, this study seeks to provide a more comprehensive understanding of how these groups operate and how their motivations influence each phase of an attack.

Therefore, this study seeks to address the following research question: What motivations prompt FIN7 to conduct cyber operations, and how do their activities align with the phases of a well-organized cyber operation?

Based on the research question, this study proposes the hypothesis that FIN7 conducts cyber operations using a structured, multi-phase approach that aligns with the model of a well-organized cyber operation, with financial gain serving as the primary motivating factor influencing each phase.

This hypothesis is evaluated by systematically mapping documented FIN7 activities to each phase of the cyber operation model and analyzing whether the group demonstrates consistent goals and objectives throughout the attack lifecycle.

Table 1. MITRE ATT&CK Group: FIN7 [2].

ID	Name	Associated Groups	Description
G0046	FIN7	GOLD NIAGARA, ITG14, Carbon Spider, ELBRUS, Sangria Tempest	FIN7 is a financially motivated threat group active since 2013 that targets industries such as retail, hospitality, financial services, and healthcare in the United States. The group has used point-of-sale malware and

			phishing campaigns to steal financial data. It has also operated through front companies and shifted toward ransomware operations, including big game hunting and Ransomware-as-a-Service (RaaS).
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Methodology

This study uses a case study research strategy to examine the cyber operations of the FIN7 threat group. The case study approach focuses on a single unit of analysis, allowing for an in-depth investigation of FIN7's behavior within a real-world context. This method is appropriate because it enables a detailed understanding of how and why FIN7 conducts its cyber operations.

The data generation method used in this study is document analysis, which involves examining publicly available sources such as the MITRE ATT&CK database, cybersecurity reports, and relevant literature. These sources provide documented evidence of FIN7's tactics, techniques, and procedures (TTPs), allowing for a structured analysis of its activities. This approach ensures that the analysis is based on verified and publicly reported evidence, increasing the reliability of the findings.

This study is guided by the following research question: What motivations prompt FIN7 to conduct cyber operations, and how do their activities align with the phases of a well-organized cyber operation? Based on this question, the study assumes that FIN7 follows a structured operational model similar to traditional Advanced Persistent Threat groups.

The analysis is organized using the seven phases of a well-organized cyber operation:

1. Target Identification
2. Reconnaissance
3. Gaining Access
4. Hiding Presence
5. Establishing Persistence
6. Execution
7. Assessment

Each phase is examined using evidence from the selected sources to determine how FIN7 conducts its operations and to evaluate whether its activities align with a structured cyber operation model.

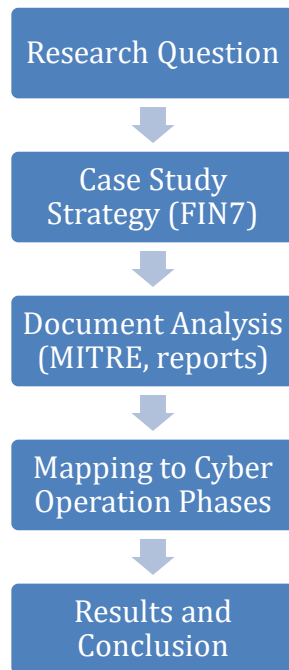


Figure 1. Case Study Research Process (Author Created).

Results

The results of this study are based on the analysis of FIN7's documented activities and how they align with the seven phases of a well-organized cyber operation. Evidence from the MITRE ATT&CK framework and related cybersecurity reports indicates that FIN7 follows a structured and methodical approach to conducting cyber operations, despite being financially motivated. Each phase is reflected in the group's tactics, techniques, and procedures.

1. Target Identification

FIN7 primarily targets organizations that process large volumes of financial transactions, including retail, hospitality, and financial service companies. These targets are selected based on their potential to provide access to payment card data and other valuable financial information. This indicates that FIN7's primary motivation is financial gain, as they focus on industries where stolen data can be easily monetized.

The primary goal of this phase is to identify targets that offer the highest potential financial return. FIN7's objective is to prioritize organizations with access to large volumes of payment data, ensuring that subsequent phases of the operation yield maximum profit.

2. Reconnaissance

During the reconnaissance phase, FIN7 gathers information about its targets to identify vulnerabilities and potential entry points. This often includes researching company structures,

employee roles, and email formats. The group uses this information to craft targeted phishing campaigns, increasing the likelihood of successful initial access.

This phase is critical to FIN7's success, as effective reconnaissance allows the group to tailor its attacks and reduce the likelihood of failure. By investing time in understanding their targets, FIN7 increases the precision and effectiveness of its operations.

In this phase, the primary objective is to gather sufficient information to enable a successful intrusion. FIN7's objective is to reduce uncertainty by identifying vulnerabilities, organizational structures, and employee behavior patterns that can be exploited during the gaining access phase.

3. Gaining Access

FIN7 commonly gains access to target systems through phishing emails that deliver malicious attachments or links. These emails are often carefully designed to appear legitimate, tricking employees into executing malware. In some cases, FIN7 has also exploited known vulnerabilities to gain entry into networks. This phase demonstrates the group's ability to use both social engineering and technical methods to achieve initial compromise.

FIN7's phishing campaigns are often highly targeted and tailored to specific organizations, a technique commonly referred to as spear-phishing. These emails frequently impersonate trusted entities or internal communications, increasing the likelihood that recipients will interact with malicious attachments or links. This level of customization demonstrates a significant investment in reconnaissance and preparation prior to execution.

In addition to phishing, FIN7 has been associated with the use of exploit kits and compromised websites to deliver malware. These techniques allow the group to broaden its attack surface and target multiple victims simultaneously, increasing the efficiency and scale of its operations.

At this stage, the objective is to successfully compromise the target system. FIN7's objective is to use efficient and scalable methods, such as phishing and exploitation, to gain initial access while minimizing the risk of detection.

4. Hiding Presence

After gaining access, FIN7 employs techniques to remain undetected within the victim's network. This includes using legitimate administrative tools and obfuscating malicious activity to blend in with normal system operations. By avoiding detection, FIN7 is able to maintain access for extended periods of time.

FIN7's use of legitimate administrative tools, often referred to as "living off the land" techniques, allows the group to evade traditional security detection mechanisms. By leveraging tools that are already present within the system, such as PowerShell or remote management utilities, FIN7 minimizes the presence of suspicious or unknown software.

This approach not only reduces the likelihood of detection but also complicates incident response efforts, as security teams may struggle to distinguish between normal administrative activity and malicious behavior.

The objective during this phase is to remain undetected within the target environment. FIN7 aims to blend malicious activity with legitimate system processes, ensuring continued access without triggering security alerts.

5. Establishing Persistence

FIN7 establishes persistence by maintaining long-term access to compromised systems. This is often achieved through the use of backdoors, compromised credentials, or scheduled tasks that allow the group to regain access even if initial entry points are discovered. Persistence enables FIN7 to continue its operations without interruption.

Maintaining persistence is essential for FIN7 because it allows the group to operate over extended periods without needing to re-establish access. This long-term presence increases the amount of data that can be collected and maximizes financial return from each compromised system.

The purpose of this phase is to maintain long-term access to the compromised system. FIN7's objective is to ensure that access can be re-established even if initial entry points are discovered and removed.

6. Execution

The execution phase involves the primary objective of FIN7's operations: the theft of financial data. The group deploys malware designed to capture payment card information and other sensitive data. In more recent operations, FIN7 has also engaged in ransomware attacks and big game hunting strategies, demonstrating an evolution in their approach to maximizing financial gain.

In addition to data theft, FIN7 has demonstrated a shift toward ransomware operations, including participation in Ransomware-as-a-Service (RaaS) models. This evolution reflects a strategic adaptation to changes in the cyber threat landscape, where ransomware attacks often yield higher and more immediate financial returns.

The group's ability to transition between different forms of cybercrime highlights its flexibility and business-like operational model, further reinforcing the idea that FIN7 operates with a level of sophistication comparable to traditional APT groups.

The execution phase focuses on achieving the primary objective of the operation. For FIN7, this objective is the extraction of financial data or deployment of ransomware to generate profit.

7. Assessment

In the final phase, FIN7 evaluates the success of its operations based on the amount of data exfiltrated or financial gain achieved. This may include analyzing stolen data, determining its value, and refining future attack strategies. This phase reflects a feedback loop that allows the group to improve the effectiveness of its operations over time.

In the final phase, the focus shifts to evaluating the success of the operation. FIN7's objective is to analyze outcomes, refine techniques, and improve the efficiency of future attacks.

Implications for Cybersecurity Defense

The analysis of FIN7's operations highlights several important implications for cybersecurity defense. Because FIN7 follows a structured and methodical approach similar to traditional APT groups, organizations must adopt equally structured defense strategies. This includes implementing layered security controls that address each phase of a cyber operation rather than focusing solely on perimeter defenses.

For example, during the target identification and reconnaissance phases, organizations can reduce risk by limiting publicly available information and conducting regular security awareness training to prevent social engineering attacks. Since FIN7 relies heavily on phishing campaigns, employee training and email filtering technologies play a critical role in disrupting the attack lifecycle before initial access is achieved.

In later phases, such as persistence and hiding presence, organizations must rely on advanced detection mechanisms, including endpoint detection and response (EDR) tools and continuous monitoring. FIN7's use of legitimate administrative tools makes detection more challenging, requiring behavioral analysis rather than simple signature-based detection methods.

Additionally, the group's shift toward ransomware and large-scale financial attacks demonstrates the importance of incident response planning and data backup strategies. Organizations must be prepared not only to detect intrusions but also to respond quickly and recover from attacks in order to minimize financial and operational impact.

Overall, the findings of this study emphasize that defending against financially motivated threat groups requires a comprehensive and proactive cybersecurity strategy that addresses all phases of a well-organized cyber operation.

Another important implication of FIN7's operational model is the increasing professionalization of cybercriminal activity. Unlike traditional cybercrime groups that operate in a more ad hoc manner, FIN7 demonstrates characteristics similar to a structured organization, including specialization of roles, coordinated campaigns, and the use of front companies to support operations. This level of organization allows the group to scale its activities and maintain long-term campaigns across multiple targets and industries.

The use of front companies and legitimate business structures also complicates attribution and law enforcement efforts. By blending malicious activity with legitimate operations, FIN7 is able to obscure its true identity and reduce the risk of disruption. This highlights the need for stronger

collaboration between cybersecurity professionals, law enforcement agencies, and private sector organizations to effectively identify and dismantle such groups.

Furthermore, FIN7's evolution toward ransomware and Ransomware-as-a-Service models demonstrates how cybercriminal groups adapt to maximize profitability. This shift indicates that future threats will likely continue to focus on scalable and high-impact attack methods. As a result, organizations must prioritize resilience, including incident response planning, regular system backups, and rapid recovery capabilities, in addition to traditional preventative measures.

Limitations of the Study

While this study provides a detailed analysis of FIN7's cyber operations, it is limited by its reliance on publicly available data. The MITRE ATT&CK framework and other open-source reports represent only a subset of the group's actual activities, meaning that some tactics and techniques may not be fully documented.

Additionally, the use of a case study approach focuses on a single threat group, which may limit the generalizability of the findings to other Advanced Persistent Threat actors. While FIN7 demonstrates characteristics of a well-organized cyber operation, other groups may operate differently based on their motivations or resources.

Despite these limitations, the study provides valuable insight into the structure and behavior of financially motivated cybercriminal organizations and contributes to a broader understanding of modern cyber threats.

Conclusion

This study examined the financially motivated cyber operations of the FIN7 threat group and evaluated whether its activities align with the phases of a well-organized cyber operation. Through a case study approach and document analysis of publicly available sources, the findings demonstrate that FIN7 follows a structured and methodical process consistent with established cyber operation models.

Each of the seven phases—target identification, reconnaissance, gaining access, hiding presence, establishing persistence, execution, and assessment—was clearly reflected in FIN7's documented behavior. The group systematically selects targets with high financial value, uses reconnaissance to identify vulnerabilities, gains access through phishing and exploitation techniques, and maintains persistence while avoiding detection. Ultimately, FIN7 executes operations focused on data exfiltration and financial gain, followed by assessing the effectiveness of their attacks to improve future operations.

The results support the conclusion that financially motivated threat actors can operate with the same level of organization and sophistication as traditional nation-state APT groups. FIN7 demonstrates that financial incentives alone are sufficient to drive complex and well-structured cyber operations.

The structured nature of FIN7's operations also suggests that future cyber threats will continue to evolve toward more organized and scalable models. As cybercriminal groups adopt increasingly sophisticated techniques, the gap between financially motivated actors and nation-state threats will continue to narrow.

Therefore, this research supports the hypothesis that FIN7 conducts cyber operations aligned with the phases of a well-organized cyber operation. These results emphasize the need for comprehensive security strategies that address both financially motivated and state-sponsored threats. Organizations must adopt proactive and layered defenses to effectively mitigate these evolving risks.

References

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